

MARK ROSENTHAL

Staff Quality Architect | Automation & Developer Platform Lead

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CORE STACK: PLAYWRIGHT | CYPRESS | JavaScript | TypeScript | Angular | React

Staff Quality Architect and Developer Infrastructure Lead with 9 years of experience engineering scalable test platforms, modular developer tooling, and deterministic CI/CD pipelines across high-traffic distributed systems. Extensive track record leading automation architecture across web, mobile, API, and OTT streaming runtimes, modernizing legacy frameworks into modern Playwright and Cypress stacks, and cutting regression cycles from two weeks to under twenty-four hours. Proven ability to rapidly prototype, scaffold, and deploy full-stack engineering utilities, harnessing agentic AI workflows and deterministic runtime guardrails to eliminate flaky test loops and accelerate developer velocity across multi-pod organizations.

IMPACT SNAPSHOT

**14X FASTER
REGRESSION**
2 weeks to 1 day

Stable E2E automation +
deterministic CI execution

**60%+ FLAKINESS
REDUCTION**

Custom CI retry logic

Intelligent retry logic +
selector stability strategy

**15-30% INFRA COST
SAVINGS**

k6 load testing strategy

Optimized load testing +
resource-efficient execution

**50+ ENG-HOURS/MONTH
SAVED**

*Reusable test architecture
+ shared automation*

Cross-suite Cypress/Playwright
frameworks

**5-6X FASTER
ONBOARDING**

*Shared test infrastructure +
reusable automation primitives*

6 weeks → under 1 week

PROFESSIONAL EXPERIENCE

TestGhost | Independent QA Automation Contractor | Remote

FEB 2025 – MAY 2026

Independent QA automation R&D project focused on Playwright/Cypress workflows, AI-assisted test planning, selector strategy, and automation migration concepts.

- Delivered contract QA automation support across Playwright/Cypress workflows, framework setup, selector stability, regression planning, and CI/CD integration.
- Created reusable QA playbooks and automation patterns for release validation, exploratory testing, regression coverage, and shift-left testing.
- Prototyped internal tooling for DOM inspection, automation scaffolding, selector strategy, and legacy test migration planning.

Generative AI for Quality Engineering

- Built GenAI-powered QA workflows using prompt chaining and RAG-style retrieval over project documentation and code artifacts to generate test scenarios, automation suggestions, and release-readiness insights.
- Designed agent-style QA workflows that break larger testing goals into chained steps, including context gathering, risk analysis, test generation, and suggested automation coverage.
- Developed AI-assisted QA tooling concepts for test planning, defect analysis, selector strategy, and automation scaffolding.
- Applied GenAI to regression planning, PR review support, exploratory testing, risk analysis, and release-quality summaries.

FloSports | QA Engineer / Automation Architect | Austin, TX

MAY 2016 - FEB 2025

Primary QA owner for automation, release validation, and quality strategy across web, mobile, API, and OTT platforms. Built scalable test frameworks, supported major platform migrations, defined release gates across engineering pods, and partnered with local/offshore teams to improve regression speed, test stability, release confidence, and QA onboarding.

- Architected enterprise-scale automation infrastructure within a complex Nx monorepo supporting distributed microservices across web, mobile, API, and OTT platforms.
- Slashed regression cycles from 2 weeks to 1 day by implementing a deterministic E2E suite and enforcing global data-tagging standards across engineering pods.
- Led QA strategy for MVP launches and high-value product initiatives, defining test scope, release gates, risk coverage, and sign-off criteria across engineering pods.
- Led quality strategy for a company-wide migration from an Angular monolith to a React micro-frontend architecture while maintaining 100% feature parity.
- Migrated legacy Java/Selenium automation to a modern Playwright/TypeScript framework, improving maintainability, observability, and developer velocity.
- Engineered a proactive k6 load testing strategy that identified backend bottlenecks and contributed to a 15–30% reduction in annual cloud infrastructure costs.
- Founded the Shift Left Guild to embed quality engineering into early SDLC planning and reduce production defects by 60%.
- Developed QA onboarding curriculum and playbooks that shortened ramp-up time from 6 weeks to under 1 week for local and offshore teams.

TECHNICAL R&D & INNOVATION | BUILDGHOST (2025-PRESENT)

Building an **AI-driven QA tooling platform** focused on automation acceleration, migrations upport, and system-level validation under real-world conditions.

- Designing a desktop-native QA automation suite in Electron with Node.js integration for Chromium-based test execution, DOM inspection, and automation workflow control.
- Building tooling for legacy test migration, including Java/Selenium-to-Playwright planning, selectoranalysis, and framework scaffolding.
- Creating IPC-based workflows to move DOM, runtime, and session context between browser views, injected scripts, and backend automation services.
- Developing selector-resilience patterns for DOM discovery, element tagging, and self-healing automation.
- Prototyping AI-assisted workflows for test migration, automation scaffolding, regression planning, and release-readiness analysis.
- Building validation tooling to test system behavior under latency, packet loss, and unstable runtime conditions.

TECHNICAL SKILLS

Automation	Cypress, Playwright, Appium, k6, Cucumber/Gherkin, Custom Test Runners, AI-Assisted Testing, Selenium WebDriver
CI/CD & DevOps	GitHub Actions, CircleCI, Jenkins, Docker, Kubernetes exposure, Terraform exposure, AWS (Lambda, S3, EC2), GCP
API & Data	Postman, REST Assured, Supertest, GraphQL, JSON Schema Validation, API Contract Testing, Data Injection
Languages	JavaScript, TypeScript, Node.js, Python, Java, SQL
Observability	Datadog, Sentry, Segment
Test Management	TestRail, Zephyr, X-Ray, JIRA administration, Confluence
Methodologies	Shift Left, Agile/Scrum, Manual/Exploratory, Regression, UAT, A/B, Smoke/Sanity, Black/White-box, CI/CD, Kanban
AI / GenAI	Prompt chaining, RAG-style retrieval, AI-assisted test generation, agent-style QA workflows, automation scaffolding

APPLIED AI ENGINEERING & RAPID PROTOTYPING

Accelerating full-stack system architecture by pairing agentic AI code synthesis with deterministic verification pipelines, enabling rapid end-to-end prototyping of desktop runtimes, CLI utilities, and distributed test harnesses with zero compromise on code quality or stability.

PRIOR PROFESSIONAL EXPERIENCE (2007 – 2016)

Neighbors Emergency / Seton | ER Technician & Lab Compliance Officer

- Served as a FEMA point of contact during mass casualty and public health events, supporting coordinated response efforts across clinical and emergency systems.
- Operated in Level 1 Trauma environments during high-impact incidents and outbreaks, including pandemic response scenarios, where precision, speed, and system reliability were critical.
- Maintained compliance across 30 plus diagnostic platforms under CLIA and COLA standards, ensuring hardware readiness, accurate data logging, and zero downtime in life critical conditions.
- Developed a zero margin for error mindset that now directly informs how I design, validate, and scale high reliability systems in my engineering work.

EDUCATION

AustinCommunity College | Studies in Computer Science | 2010-2016